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A robust deep learning algorithm for lung cancer detection from computed tomography images

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ABSTRACT

Detecting lung cancer at its earliest stage offers the best possibility for a cure. Chest computed tomography (CT) scans are a valuable tool for early diagnosis. However, the initial stages of lung cancer may present patterns in the images that are not easily detectable by radiologist, potentially leading to misdiagnosis. Although automated approaches using deep learning (DL) algorithms have been proposed, it depends on a substantial amount of data to achieve diagnostic accuracy comparable to that of radiologists. To alleviate this challenge, this study proposes a DL algorithm that uses an ensemble of convolutional neural networks and trained on relatively small dataset (IQ_OTH/NCCD dataset) to automate lung cancer diagnosis from patient chest CT scans. The method achieved an accuracy of 98.17 %, a sensitivity of 98.21 %, and a specificity of 98.13 % when categorizing scans as either cancerous or non-cancerous. Similarly, it achieved an accuracy of 95.43 %, a sensitivity of 93.40 %, and a specificity of 97.09 % when classifying scans as normal or containing benign or malignant pulmonary nodules. These results demonstrate superior performance compared to previously proposed models, highlighting the effectiveness of DL algorithms for early lung cancer diagnosis and providing a valuable tool to assist radiologists in their assesments.

1. Introduction

Lung cancer is a deadly disease and the leading cause of cancer-related deaths worldwide [1]. Early detection can favor prognosis and enhances the quality of life for patients [2]. However, many lung cancer cases are identified at advanced stages when the disease has already spread to other parts of the body [3]. Statistics indicate that about 55 % of patients diagnosed with lung cancer at an early stage survive for more than five years, compared to only 5 % of those diagnosed at later stages [4]. This late diagnosis severely limits the chances of patient survival [5].

Medical images, particularly chest computed tomography (CT) scans provides a robust tool for radiologist to diagnose lung cancer [6]. Analyzing and interpreting these images offers a non-invasive method for detecting the disease in its early stages [7]. However, manually analyzing CT scans is a complex task that requires years of clinical experience [8]. Furthermore, the onset of lung cancer may present patterns that are not visible to the human eye, which can lead to the oversight of significant information and result in potential misdiagnosis [9,10]. Additionally, the manual analysis of medical images is time-consuming and can overburden radiologists, resulting in prolonged waiting times for patients in hospitals [8].

Recently, deep learning (DL), a branch of artificial intelligence (AI), has shown promise in the early diagnosis of lung cancer using patient CT scans [11]. DL algorithms use artificial neurons to extract important distinguishing features from data, and their performance can be enhanced through systematic training [12]. Additionally, DL algorithms can process larger volumes of data more quickly than radiologists can [13]. The DL methods have been applied to automate detection, classification, and segmentation of pulmonary nodules in CT scans [14,15]. Furthermore, numerous studies have demonstrated that DL models can outperform radiologists in interpreting patient CT scans [16,17].

Despite the success of deep learning in detecting pulmonary nodules from CT images, it depends on a substantial amount of data to achieve

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diagnostic accuracy comparable to that of radiologists, yet the availability of clinical training data is limited [17,18]. This limitation creates a gap between DL research and its clinical applications [19,20]. To address this challenge, this study aims to develop and train an automated detection system that would be effective at detecting malignancy in lung CT scans using a relatively small dataset. Additionally, the study provides an overview of pulmonary nodule detection from patient CT scans using convolutional neural networks (CNN).

The rest of the paper is organized as follows. A review of related literature is presented in Section 2. Materials and Methodology is presented in Section 3. The study results are presented in Section 4. Section 5 discusses the study results. Finally, Section 6 provides the concluding remarks and future work.

2. Literature review

Deep learning gained significant popularity in 2012 when Krizhevsky et al. [21] successfully trained a CNN to classify approximately 1, 000,000 images across 1000 different categories. This breakthrough led to the development of several CNN architectures that have excelled in various visual tasks such as object detection, classification, and segmentation. CNNs are at the forefront of visual modelling, and numerous studies have extended their applications to the diagnosis of lung cancer [13,15].

In 2015, Hua et al. successfully trained a CNN on the Lung Image Database Consortium (LIDC-IDRI) dataset to classify pulmonary nodules, achieving a specificity of 78.70 % and a sensitivity of 73.30 % [22]. The LIDC-IDRI dataset is a collection of lung images compiled by the National Cancer Institute, which includes 1018 helical thoracic CT scans obtained from 1010 unique patients [23]. Additionally, this dataset contains an XML file documenting the outcomes of a two-phase image annotation process conducted by four experienced thoracic radiologists. The creation of this dataset and the success of the study paved the way for additional chest CT datasets and the development of several deep learning models that utilize these datasets to diagnose lung cancer.

In 2016, the LUNA16 subset was created from the LIDC-IDRI dataset. This subset excludes scans with a slice thickness greater than 2.5 mm, resulting in a total of 888 CT scans [24]. The LUNA16 dataset includes 1186 nodules along with their corresponding segmentation masks. Furthermore, several other datasets have been introduced, including the National Cancer Institute (NLST) dataset [25], the Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases (IQ-OTH/NCCD) dataset [26], the Ali-Tianchi dataset [27], the Danish Lung Cancer Screening Trial (DLCST) dataset [28], and the Chest CT scan dataset [29].

2.1. Pulmonary nodule detection using convolutional neural Networks from computed tomography scans

In 2017, Shen et al. [30] developed a multi-crop 2D CNN model to classify pulmonary nodules as either malignant or non-malignant and to predict their size. They employed a multi-crop pooling strategy to extract different regions from the image feature map and utilized repeated max pooling to gather nodule characteristics, thus eliminating the need for nodule segmentation and handcrafted feature extraction. For their study, the authors used the LIDC-IDRI dataset, which includes CT images of nodules ranging from 3 mm to 30 mm in diameter. They reported an accuracy of 87.14 %, a sensitivity of 77.00 %, a specificity of 93.00 %, and an area under the curve (AUC) of 93.00 %. In the same year, Ding et al. [31] used an ensemble of 3D CNNs to detect malignancy in CT images. They implemented a deconvolutional algorithm in conjunction with a Faster Region-based CNN to identify suspicious nodules within the CT slices. Following this, a 3D CNN was employed to reduce false positive nodules. Candidate nodules were further processed using three 3D attention-based CNNs to determine which were malignant. This approach achieved a detection sensitivity of 94.6 % at 15.0 false positives per scan on the LUNA16 dataset.

In 2018, Liu et al. [32] developed a 2D convolutional binary-tree network called 'DenseBTNet.' The authors enhanced the traditional DenseNet by incorporating a center crop algorithm and reconfiguring its architecture. They split and merged the transition layers with dense blocks and adjusted the feature-map transition mode to streamline the model while improving multi-scale feature extraction. This approach achieved an accuracy of 88.31 % and an AUC of 93.15 % on the LIDC-IDRI dataset, surpassing DenseNet's performance. Jin et al. [33] employed a 3D CNN with residual connections to predict lung cancer in chest CT scans. They employed pooling, cropping, and online hard sample selection techniques to identify false positive nodules, which enhanced the model's prediction accuracy. Their method achieved an accuracy of 80.2 % at 0.125 false positives per scan and 94.1 % accuracy at 8 false positives per scan. Zhu et al. [34] introduced a two-component 3D dual-path network model for diagnosing lung cancer from chest CT scans. The nodule detection module features a 3D Faster Region-based CNN with 3D dual-path blocks that mimic a U-net architecture. The nodule classification module employs a Gradient Boosting Machine that incorporates the features from the dual-path network. This method achieved an accuracy of 92.74 % on the LIDC-IDRI dataset, outperforming the results obtained by four physicians. In the same year, George et al. [35] proposed 'DetectNet,' a system based on the You Only Look Once (YOLO) method for detecting and categorizing pulmonary nodules in CT scans as either cancerous or non-cancerous. Their system achieved a sensitivity of 89.00 %, with an average of 6 false positives per scan on the LIDC-IDRI dataset.

In 2019, Xie et al. [36] developed a framework that included two region proposal networks, a deconvolutional layer, and a 2D CNN to diagnose lung cancer using patient CT scans. The authors trained three separate models to classify detected nodules as either cancerous or non-cancerous. They evaluated their method using the LUNA16 dataset and reported a sensitivity of 86.42 % for nodule detection. Liu et al. [37] used a 3D feature pyramid network to detect and locate pulmonary nodules in CT scans. This approach utilized multi-scale features to enhance the resolution of nodules from the original CT images and employed a parallel top-down strategy to merge high-level semantic features with low-level features. Additionally, they introduced a network called 'HS2' to analyze appearance differences across continuous CT slices, which helped select potential nodule candidates and reduce false positive predictions. Their method achieved a sensitivity of 90.4 % with 0.125 false positives per scan on the LUNA16 dataset. In the same year, Ardila et al. [16] used a combination of three networks to predict the risk of lung cancer by analyzing both current and previous CT scans of patients. The first network was a 3D CNN trained with a focal loss function for whole image volume analysis. The second network identified potential cancerous regions of interest from the CT scans, while the third CNN predicted the probabilities of malignancy based on the representations provided by the first two networks. This method achieved an area under the receiver operating characteristic (AUROC) of 94.40 % using data from 6716 cases in the National Cancer Institute (NCI) screening trial, surpassing the results obtained by six radiologists.

In 2020, Lyu et al. [38] utilized a multi-level cross-residual 2D CNN to categorize patient CT scans into three categories: benign, indeterminate, and malignant. They also performed a two-class classification to determine whether the scans contained benign or malignant pulmonary nodules. Unlike traditional ResNet architectures, where layers are connected in a linear fashion, the residual modules in their network were interconnected in a crossover manner. This approach achieved an accuracy of 85.88 % for the three-class classification and 92.19 % for the two-class classification on the LIDC-IDRI dataset, surpassing ResNet-50. Yu et al. [39] employed an improved 2D VGG and ResNet architecture to identify potential nodule candidates and classify the detected nodules as either cancerous or non-cancerous. They utilized techniques such as histogram equalization, binarization and region growing to segment the lung parenchyma from the CT scans. Their method achieved a detection rate of 90.9 % on the LIDC-IDRI dataset. El-Bana et al. [40] introduced a method using DeepLab for the semantic segmentation of lung parenchyma from CT scans. The segmented lung parenchyma was then used to train a faster region-based CNN and a single-shot detector with Inception-V2 as the backbone. This approach resulted in a detection sensitivity of 96.4 % on the LUNA16 dataset. In the same year, Cai et al. [41] implemented a 3D CNN to detect pulmonary nodules and generate 3D visualizations of the detected nodules. They employed a ray-casting volume rendering algorithm to create 3D representations of the nodules. Additionally, a mask region-CNN with a ResNet-50 backbone and a feature pyramid network was used to generate multiscale feature maps, while a regional proposal network was utilized to propose bounding boxes around potential nodule candidates. This method achieved accuracies of 88.1 % and 88.7 % at 1 and 4 false positives per scan, respectively, for the Ali-Tianchi and LUNA16 datasets.

In 2021, Jena et al. [42] incorporated a Gaussian mixture model into a region-based 2D CNN to identify cancerous regions in patient CT scans. The CT images from the LIDC-IDRI dataset were preprocessed using Wiener and Gaussian filters to reduce noise, and a region-growing segmentation technique was employed to isolate the lung region of interest. This method achieved a classification accuracy of 87.79 % on the LIDC-IDRI dataset. Surendar et al. [43] utilized a hybrid classifier that combined a 2D CNN with an adaptive sine-cosine crow search algorithm to detect lung cancer in CT images. For image preprocessing, they used a non-local means algorithm along with Masi entropy-based multilevel thresholding coupled with a salp swarm algorithm to segment the lung nodule region. This approach achieved an accuracy of 99.17 % on the LIDC-IDRI dataset. Salama et al. [44] employed a U-Net and ResNet-50 architecture for lung cancer diagnosis. The U-Net was used to segment the lung parenchyma from the CT scans, while the ResNet-50 determined the malignancy of the nodules. The authors applied an image-sized dependent normalization technique and utilized a Wiener filter to process the images before passing them to the CNNs. Their model was evaluated using the NLST and LUNA16 datasets, achieving an accuracy of 98.98 %, sensitivity of 98.99 %, and AUC of 98.65 %. AL-Huseiny et al. [45] adopted a modified 2D GoogLeNet with transfer learning approach to classify patient CT scans as either containing malignant or non-malignant pulmonary nodules. They used Gabor filtering as well as image dilation and erosion techniques to segment the lung parenchyma from the CT scans. Additionally, bounding boxes were introduced around the segmented lungs to define a rectangular region of interest. This method achieved a classification accuracy of 94.38 %, specificity of 93.7 %, and sensitivity of 95.08 % on the IQ_OTH/NCCD dataset. Zheng et al. [46] developed a two-stage framework for detecting pulmonary nodules in CT scans. The first stage employs a multiplanar nodule detection system that is trained using axial, coronal, and sagittal CT slices. The second stage involves a multiscale false positive reduction module, which uses a 3D multiscale dense convolutional neural network (CNN) to eliminate non-nodules. The method was evaluated on the LIDC-IDRI dataset and achieved a sensitivity of 94.2 % at 1.0 false positive (FP) per scan and 96.0 % at 2.0 FPs per scan for nodules larger than 6 mm. For nodules smaller than 6 mm, the sensitivity was 93.4 % at 1.0 FP per scan and 95.0 % at 2.0 FPs per scan.

In 2022, Chi et al. [47] introduced an inception module and skip connections within a U-Net architecture to perform lung segmentation from CT scans. They employed a dilated convolution in a second U-Net to detect suspicious nodules, while a third U-Net, utilizing multi-resolution and multi-scale pooling convolution, identified true nodules. This method achieved an accuracy of 94.75 %, a sensitivity of 90.36 %, and a specificity of 96.55 % on the LUNA16 dataset. On the Ali-Tianchi dataset, it achieved 93.90 % accuracy, 89.88 % sensitivity, and 94.76 % specificity. In another study, Solyman et al. [48] applied both transfer learning and ensemble learning techniques to classify CT scans into three categories: normal, benign, or malignant pulmonary nodules. Their single VGG 2D CNN model with transfer learning achieved 89.00 % accuracy, while the ensemble model which included VGG-16, ResNet50, InceptionV3, and EfficientNetB7 achieved 92.80 % accuracy on the IQ-OTH/NCCD dataset. Additionally, Aslani et al. [49] employed a 3D CNN with a time-series approach to detect lung cancer from CT images. Their model integrated data from three longitudinal domains: nodule-specific, lung-specific, and clinical demographic data. For nodule detection, they used a 3D U-Net encoder-decoder architecture with ResNet10 as the encoder backbone. The decoder featured a mirrored architecture of the encoder with reduced convolutional blocks. They incorporated squeeze-and-excitation blocks to enhance the contextual feature extraction from the CT images. To automate the region proposals, a 3D region proposal network was utilized. To extract nodule-level features from longitudinal inputs, a long short-term memory (LSTM) layer was implemented. For predicting malignancy scores, the authors employed a ResNet-10 that combined nodule-level and lung-level features. This method achieved an AUC of 88.00 % on the NLST dataset.

In 2023, Jain et al. [50] developed a two-stage 2D CNN for detecting pulmonary nodules from CT scans. The first stage employs a U-Net for nodule segmentation, while the second stage uses a CNN to classify the nodules. This method achieved an accuracy of 84.40 % on the LIDC-IDRI dataset. Shah et al. [51] utilized an ensemble of three 2D CNNs with different layer configurations and pooling techniques to diagnose lung cancer. They incorporated an image preprocessing pipeline that crops the CT images around the nodule coordinates to enhance computational efficiency. This approach achieved an accuracy of 95.00 % on the LUNA16 dataset. Lin et al. [52] implemented a 3D framework called IR-UNet++ for the automated detection of pulmonary nodules from CT scans. This framework combined Inception Net and ResNet as building blocks. In addition, they introduced a squeeze-and-excitation structure to improve feature extraction. Furthermore, they utilized two short skip pathways based on the U-Net architecture to enhance overall system performance. A thresholding algorithm was employed to segment the lung parenchyma from the CT scans. The method achieved sensitivities of 90.13 %, 94.77 %, and 95.78 % at 1 FP/scan, 4 FPs/scan, and 8 FPs/scan, respectively, on the LUNA16 dataset.

In 2024, Jian et al. [53] developed a two-branch 3D CNN named DBPNDNet to detect pulmonary nodules in CT scans. One branch was designed for extracting candidate regions for pulmonary nodule detection, while the other branch was employed for the semantic segmentation of lesion regions associated with the nodules. They implemented a 3D attention-weighted feature fusion module to enhance the performance of the detection branch by deriving relevant nodule characteristics. This method achieved a sensitivity of 97.14 % with an average of 8 false positives per scan on the LIDC-IDRI dataset. Similarly, Wu et al. [54] introduced a multi-kernel-driven 3D CNN called MK-3DCNN, which utilized a multi-mode mixed pooling strategy for the detection of pulmonary nodules in chest CT scans. The architecture was based on a residual learning-based UNet-like encoder-decoder framework, designed to capture multi-layer features, while a multi-kernel joint learning block was employed to capture 3D multi-scale spatial information of nodules in the CT scans. Lung segmentation was performed using a thresholding method combined with a convex hull algorithm. This approach achieved a sensitivity of 95.62 % with 8 false positives per scan on the LUNA16 dataset. In 2025, Abe et al. [55] introduced a cost-effective DL model to diagnose lung cancer from patient CT scans. The method utilized a five-layered CNN with Mavage pooling techniques and was trained on the IQ-OTH/NCCD dataset The method achieved an accuracy of 99.70 % and 99.53 % AUROC when classifying scans as normal or cancerous. It achieved an accuracy of 90.24 % and an AUROC of 94.63 % when the scans were classified as normal or containing benign or malignant pulmonary nodule, surpassing ResNet50 and GoogLeNet.

3. Materials and Methodology

3.1. Dataset Description

This study uses the publicly available IQ-OTH/NCCD lung cancer data [26]. The dataset is a collection of 1097 CT images obtained from a total of 110 cases at the Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases in 2019. The data includes CT images of healthy individuals and lung cancer patients with diverse demographic characteristics. The oncologists and radiologists at the collecting center classified the images into benign, malignant, and normal cases. Specifically, 15 cases were classified as benign, containing a total of 120 images; 40 cases were classified as malignant, containing 416 images; and 55 cases were classified as normal, containing 561 images. The images were initially retrieved in DICOM format and were later converted to JPG files of 512 x 512 pixels. The images in the dataset were obtained from a SOMATOM Siemens scanner with 120 kV, 1 mm slice thickness, 350 to 1200 Hounsfield window width range, and 50 to 600 window center range.

3.2. Segmentation of the lung region of interest

We used Morphological operations and crop algorithms to segment the lung regions of interest (ROI) from each patient CT scan. First, we generate a lung mask using a threshold algorithm given by Ref. [56].

$$G(x,y) = \begin{cases} 1, if(x,y) > T \\ 0, if(x,y) \leq T \end{cases}$$

Where G is the threshold function, x and y are point coordinates, 1 is object, 0 is background, and T is the threshold value.

We further employed cropping algorithms to capture tumor masks that were omitted by the threshold algorithm. The threshold and crop masks were then combined to generate a more robust lung mask. The mask was expanded by using dilation and fill-hole algorithms to create a final lung mask. Region of Interest (RoI) were then retrieved by multiplying the generated mask with the original patient scan. Fig. 1 displays examples of segmented lung RoI.

3.3. Data augmentation

We performed data augmentation to improve the scale and diversity of the dataset. We employed three data augmentation techniques: vertical flip, horizontal flip, and image rotation. Additionally, we oversampled the benign class to address the limitation of class imbalance which could potentially negatively impact model performance.

3.4. Model architecture

The proposed model, 'DeepNodule-Detect,' is an ensemble of three distinct 2D CNN models. The first model (Model 1) consists of five convolution layers with ReLu activation function. The convolution layers employ a 3 × 3 kernel size, stride and padding of 1. A 2 × 2 Maxpool layer is introduced after each convolution layer to down sample the extracted feature. The output of the last Maxpool layer is flattened and passed unto three linear layer that uses a ReLu activation. Finally, the output of the last linear layer is passed onto an output layer that uses a SoftMax activation function.

The second model (Model 2) uses a 2D convolution with a 4 × 4 kernel size, stride of 2, padding of 1 and ReLu activation function that is repeated in five steps with no Maxpool. The convolution layers perform both feature extraction and down sampling. The output of the last convolution layer is flattened and passed unto three linear layers with ReLu activation function. Finally, the output of the last linear layer is passed onto an output layer that uses a SoftMax activation function.

The third model (Model 3) uses both convolution and Maxpool layers. The first layer employs a 2D convolution with a 4 × 4 kernel size, stride of 2, padding of 1 and ReLu activation function with no Maxpool. The second block is 2D convolution with a 3 × 3 kernel size, stride and padding of 1, activation function and a 2 × 2 Maxpool layer. The third and fifth layer uses the same configuration as the first layer while the fourth layer uses the same configuration as the second layer. The convolution layers perform the feature extraction while both the convolution layers and Maxpool layers perform down sampling. The output of the last layer is flattened and passed unto three linear layers that uses ReLu activation functions. Finally, the output of the last linear layer is passed onto an output layer that uses a SoftMax activation function. Model 3 architecture is illustrated in Fig. 2.

The DeepNodule-Detect was attained through a linear ensemble of the three distinct CNN models (Models 1, 2 and 3). To achieve this, we trained the CNNs independently on the dataset, thereafter, the values of the predicted probabilities for each model were summed across the classes. Finally, the maximum value of the summed predictions was selected as the ensemble model final prediction.

3.5. Experiment settings

The experiments were conducted using PyTorch (version 1.13.0) in a Jupyter Notebook (version 6.4.12) running on an NVIDIA QUADRO RTX 3000 GPU. We applied a train-test split with a ratio of 80:20. Before segmenting the lung RoI, all images were resized to 512 x 512 pixels. During training, we set the batch size to 16 and employed a cross-entropy loss function along with gradient clipping. We used the Adam optimization algorithm and implemented a one-cycle learning rate

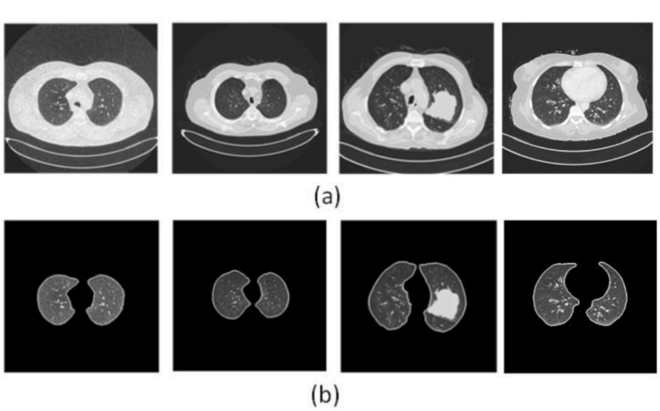

Fig. 1. Chest CT scans (a) and corresponding segmented region of interest (b).

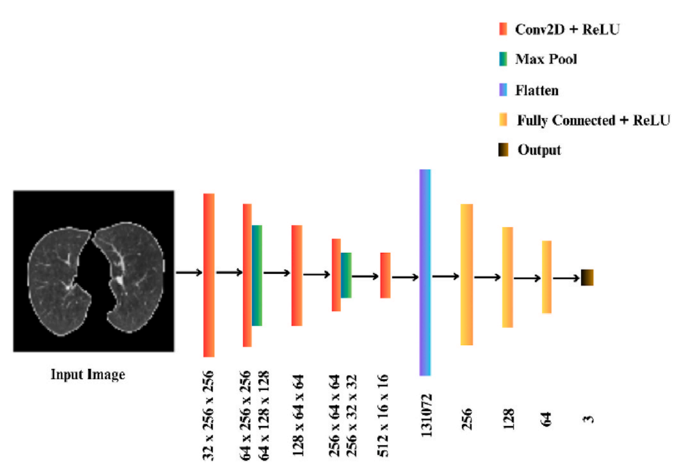

Fig. 2. Model 3 architecture.

strategy, starting at a value of 0.0001. Given the size of the training dataset, we trained the algorithm for 30 epochs to mitigate overfitting. Fig. 3 shows an Instance of the training and validation loss over 30 epochs.

3.6. Evaluation metrics

We evaluated the performance of DeepNodule-Detect by using the pathologically defined diagnosis in the dataset as our ground truth. To measure this performance, we calculated commonly used deep learning image classification metrics, including accuracy, sensitivity, and specificity. These metrics were derived from the model's predicted outcomes: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The accuracy, sensitivity, and specificity is given by Ref. [35].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

Where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

4. Results

4.1. Lung cancer detection

To detect lung cancer in patient CT scans, we employed DeepNodule-Detect to classify the scans as either cancerous or non-cancerous. Initially, we assessed the algorithm's performance without data augmentation and segmentation of the lung RoI. The results indicated an accuracy of 87.67 %, a sensitivity of 98.85 %, and a specificity of 80.30 %. After data augmentation and segmentation of the RoI was introduced, the algorithm's accuracy improved by 10.50 %, and specificity increased by 17.83 %. The results also showed that the ensemble model surpassed each individual CNN model. Fig. 4 shows the results of the confusion matrix while detailed classification results are presented in Table 1.

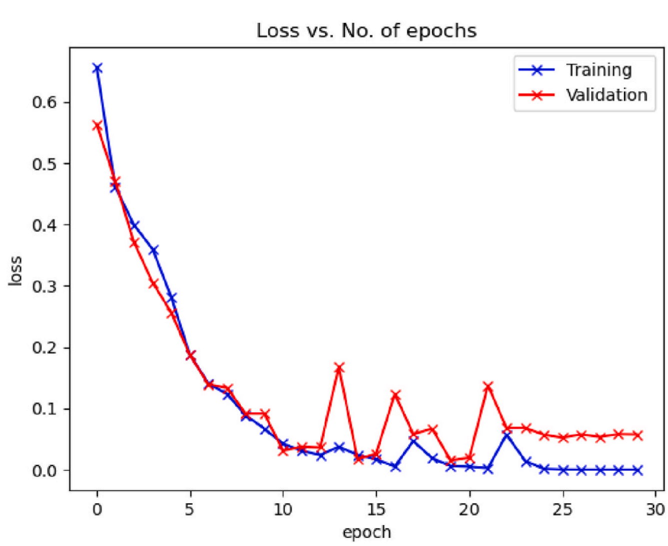

Fig. 3. DeepNodule-detect training and validation loss over 30 epochs.

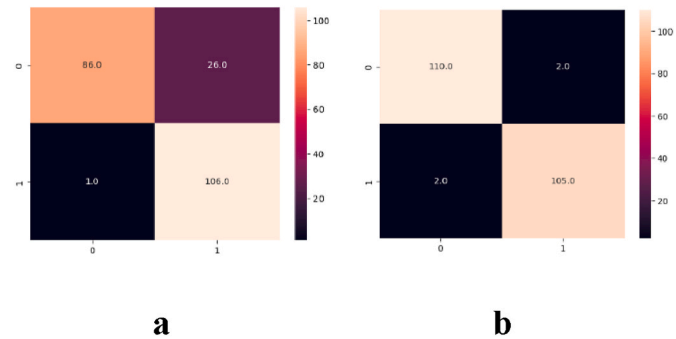

Fig. 4. Confusion matrix for DeepNodule-Detect in detecting lung cancer using the IQ-OTH/NCCD dataset. a = when the lung region of interest were not segmented, b = when the lung regions of interest were segmented, and data augmentation was employed. 0 represents scans containing malignant nodule while 1 represents normal scans.

4.2. Classification of pulmonary nodules

The DeepNodule-Detect was implemented to classify CT scans of patients as either normal or indicative of benign or pulmonary nodules. Initially, we implemented the algorithm without data augmentation and segmentation of the lung RoI. In this configuration, DeepNodule-Detect achieved an accuracy of 77.17 %, with a sensitivity of 78.70 % and a specificity of 75.68 %. When we segmented the lung RoI, the DeepNodule-Detect improved by 17.35 % accuracy, 13.63 % sensitivity and 21.35 % specificity. Furthermore, we implemented the algorithm when RoI segmentation was performed with data augmentation. The performance of the DeepNodule-Detect improved by 0.91 % accuracy, 1.03 % specificity and 0.06 % specificity. Furthermore, the study results indicate that the ensemble model outperform the individual CNN models. Fig. 5 shows the confusion matrix while detailed classification results are presented in Table 2

4.3. Comparison with previous studies

We compared the performance of DeepNodule-Detect with previous studies that analyzed the IQ-OTH/NCCD lung cancer data. In classifying CT scans as either cancerous or non-cancerous, DeepNodule-Detect outperformed the study by Al-Huseiny et al. [45], improving accuracy by 3.79 %, sensitivity by 3.13 %, and specificity by 4.43 %. For the classification of CT scans as either normal or showing benign or malignant pulmonary nodules, DeepNodule-Detect also surpassed the results from Kareem et al. [57] and Al-Yasiry et al. [58] in accuracy, sensitivity, and specificity. Detailed results are presented in Table 3.

5. Discussions

Lung cancer affects all genders, and it is responsible for the highest number of cancer-related deaths worldwide [1]. Early diagnosis is crucial for achieving a cure [2]. While chest computed tomography scans provide radiologists with a reliable tool for early detection, many lung cancer cases are diagnosed at later stages, which limits survival outcomes for patients [8,10]. Additionally, nodules detected during screening may sometimes be false positives, leading to unnecessary follow-ups and costly invasive procedures that can negatively impact a patient's emotional and mental health [3].

In this study, a deep learning approach was utilized to automate the diagnosis of lung cancer using patient chest CT scans. We employed three CNN models that were ensembled linearly to form DeepNodule-Detect. The individual CNN models were trained on the IQ_OTH/NCCD lung cancer dataset and were designed to classify CT scans as either normal or indicative of lung cancer. Additionally, they were employed to categorize patient scans into normal scans, scans

Table 1
Performance of DeepNodule-Detect in detecting lung cancer in patient computed tomography scans.

Model	Augmentation	Segmentation	Accuracy (%)	Sensitivity (%)	Specificity (%)
Model 1	X	X	82.55	95.37	80.15
Model 2	X	X	83.15	94.37	78.53
Model 3	X	X	85.27	93.29	79.55
DeepNodule-Detect	X	X	**87.67**	**98.85**	**80.30**
Model 1	✓	X	88.23	92.35	89.57
Model 2	✓	X	90.55	95.17	88.26
Model 3	✓	X	91.35	89.39	**94.75**
DeepNodule-Detect	✓	X	**93.57**	**95.67**	92.37
Model 1	X	✓	90.51	95.35	92.17
Model 2	X	✓	91.27	93.11	93.65
Model 3	X	✓	92.11	92.77	**95.21**
DeepNodule-Detect	X	✓	**94.27**	**96.27**	94.25
Model 1	✓	✓	95.51	97.73	94.32
Model 2	✓	✓	95.87	94.91	97.71
Model 3	✓	✓	96.38	96.29	95.27
DeepNodule-Detect	✓	✓	**98.17**	**98.21**	**98.13**

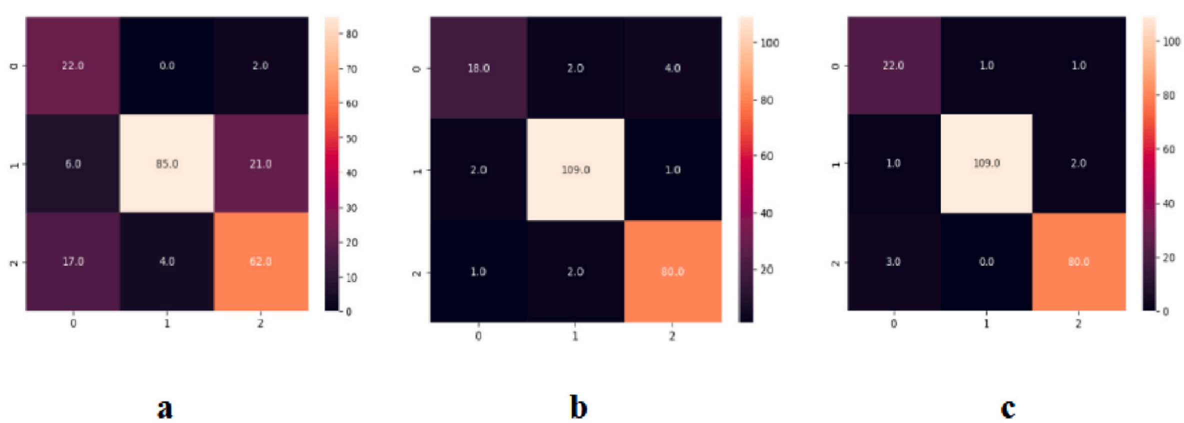

Fig. 5. Confusion matrix for DeepNodule-Detect in classifying pulmonary nodules using the IQ-OTH/NCCD dataset. a = when the lung regions of interest were not segmented, b = when the lung regions of interest were segmented, c = when the lung regions of interest were segmented, and data augmentation was employed. 0, 1 and 2 represents the benign, malignant and normal classes respectively.

Table 2
Performance of DeepNodule-Detect in classifying pulmonary in patient computed tomography scans.

Model	Augmentation	Segmentation	Accuracy (%)	Sensitivity (%)	Specificity (%)
Model 1	X	X	75.37	77.24	72.68
Model 2	X	X	74.51	75.23	**76.15**
Model 3	X	X	75.13	76.57	74.28
DeepNodule-Detect	X	X	**77.17**	**78.70**	75.68
Model 1	✓	X	87.37	90.27	92.78
Model 2	✓	X	86.68	**93.55**	91.25
Model 3	✓	X	85.11	89.57	93.67
DeepNodule-Detect	✓	X	**89.27**	91.75	**94.37**
Model 1	X	✓	92.83	91.55	96.45
Model 2	X	✓	93.28	90.28	95.68
Model 3	X	✓	92.85	91.78	94.85
DeepNodule-Detect	X	✓	**94.52**	**92.37**	**97.03**
Model 1	✓	✓	93.77	**95.28**	95.75
Model 2	✓	✓	94.68	91.58	96.86
Model 3	✓	✓	94.15	93.15	96.37
DeepNodule-Detect	✓	✓	**95.43**	93.40	**97.09**

containing benign nodules, or scans containing malignant nodules. The CNN models comprise five convolutional layers with distinct feature extraction module. Thus, providing robustness in the overall architecture. The distinct architectures also allowed for unequivocal capture of subtle differences between normal and diseased scans.

DeepNodule-Detect features a straightforward architecture. The linear ensemble is parameter learning-free resulting in minimal computational overhead. The study results showed that the ensemble model surpassed the performance of the individual CNN model demonstrating the effectiveness of the ensemble model in detecting lung cancer and classifying pulmonary nodules in patient chest CT scans. The ensemble model advances the confidence of prediction by minimizing the false positive rate. Furthermore, the ensemble method simulates combining the diagnosis of three oncologist in a clinical setting. These results support the results of previous studies demonstrating the effectiveness of using ensemble models for image classification [59,60].

Table 3
Comparison of the performance of DeepNodule-Detect with previous studies on the IQ-OTH/NCCD lung cancer data.

Author	Number of Classes	Accuracy (%)	Sensitivity (%)	Specificity (%)
Al-Huseiny et al. [45]	2	94.38	95.08	93.70
Ours	2	**98.17**	**98.21**	**98.13**
Kareem et al. [57]	3	89.89	**97.14**	**97.50**
Al-Yasriy et al. [58]	3	93.54	95.71	95.00
Ours	3	**95.43**	93.40	97.09

Chest CT scans encompass several organs beyond the lungs, including the heart, ribs, and trachea, which may adversely affect the model's performance. The presence of the CT table in the images can also contribute to noise. To address these challenges, we segmented the lung RoI from the entire CT image before inputting it into the network. The chest window in CT scans typically falls within defined Hounsfield units, which are used to map pixels into gray values [61]. This allowed us to utilize a segmentation pipeline that employs morphological operations and cropping algorithms. This approach is fast, computationally efficient, and does not require parameter training, in contrast to trainable models like U-Net [62]. Our findings indicate that this method significantly improved the network's performance in terms of accuracy, sensitivity, and specificity, thereby supporting the results of prior studies that have reported enhanced performance with segmented lung RoIs [45,58].

The IQ_OTH/NCCD lung cancer dataset is relatively small compared to other chest CT datasets, such as LUNA16 [24] and NLST [25]. Deep learning is data-intensive [18], and using a small dataset can limit performance [20]. To address this issue, we employed data augmentation techniques. We applied three affine image transformation methods: rotation, vertical flip, and horizontal flip. These techniques were chosen for their ease of implementation and their ability to preserve labels. The study results demonstrated that these techniques improved the overall performance of the DeepNodule-Detect model. With the introduction of data augmentation, the model achieved an overall accuracy of 98.17 %, a sensitivity of 98.21 %, and a specificity of 98.15 % when categorizing scans as either cancerous or non-cancerous. Furthermore, it achieved an accuracy of 98.17 %, with a sensitivity of 97.32 % and a specificity of 99.07 % when classifying scans as normal or containing benign or malignant pulmonary nodules. These results showcased superior performance compared to previous models, particularly those based on traditional machine learning approaches [57], highlighting the advantages of deep learning over these conventional methods. Additionally, the model outperformed a pre-activated GoogleNet that had been retrained using a transfer learning approach [45], demonstrating the effectiveness of custom CNNs [46,53,55] over state-of-the-art CNNs for lung cancer diagnosis from CT scans.

6. Conclusion

Deep learning algorithms can effectively detect pulmonary nodules from patient chest computed tomography scans when trained on relatively small dataset. The use of ensemble technique, segmentation of the lung regions of interest and using data augmentation can significantly improve their performance. In the future, we shall extend our approach to larger lung cancer datasets and other cancer types across multiple imaging techniques such as magnetic resonance imaging and positron emission tomography. Additionally, we shall a consider a clinical validation of the proposed method.

CRediT authorship contribution statement

A.A. Abe: Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **M. Nyathi:** Conceptualization, Project administration, Supervision, Writing – review & editing. **A.A. Okunade:** Project administration, Supervision, Writing – review & editing. **W. Pilloy:** Writing – review & editing. **B. Kgole:** Writing – review & editing. **N. Nyakale:** Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

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